1

Introduction to Logic

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Notation

$\mathcal{M}_+(X, \mathcal{A})$	finite measures on $(X, \mathcal{A})$
v	velocity

Preface

cc-by-nc-sa

Chapter 1

Logic

Chapter 2

Set Theory

2.1 Basic Structures

move to notes on intro to math and make these set theory notes assuming familiarity with producsts of two sets, similar to the set theory in the analysis notes.

In this section, we provide an exposure to a basic framework, called set theory. This framework provides a unifying language which allows us to discuss many objects of interest in mathematics by providing the ability to

1. more precisely define objects of interest,
2. more precisely formulate and prove statements about those objects
3. perform (1) and (2) in a way which is aligned with our intuition.

At an intuitive level, the main idea of a **set** is that of a collection of distinct objects, which we refer to as **elements** or **members**. When denoting a set, it is customary to list the elements of the set, separated by a comma and inside of a pair of curly brackets. Some examples of sets include the following:

- the set of letters of the English alphabet: $\{a, b, \ldots, z\}$, a set with 26 elements
- the set of inhabitants of a specific household: $\{$"Marielle", "Fred", "Dexter"$\}$, a set with 3 elements
- the set of nonnegative even integers: $\{0, 2, \ldots\}$, a set with infintely many elements

Note 2.1.0.1. Since the elements of a set must be distinct, we would consider, for instance, the sets $\{a, b, c\}$ and $\{a, a, b, c\}$ to represent the same set.

When expressing that a given object is a member of set, it is customary to use the **membership symbol**, denoted $\in$. Some examples include the following:

- $2 \in \{1, 2, 3, 4\}$
- "Earth" $\in \{$"Mercury", "Venus", "Earth", "Mars"$\}$.

Note 2.1.0.2. We note that this introduction to set theory is not formal, but intuitve. As such, we make no attempt to precisely define the notion of a set or the notion of membership to a set.

At this point in our discussion, it might seem that any advantage conferred by the language of sets in studying mathematics remains to be seen. Indeed, more time and experience may be required to fully appreciate the power of this framework, but for now we summarize why set theory is relevant in mathematics with the following mantra:

In mathematics, everything is a set.

There is a similar perspective adopted in the context of Linux operating systems where every object is a file or in the context of object oriented programming where every object is a class.

If one already has familiarity with various objects in mathematics such as various numbers, functions, and shapes, they might wonder how these objects could be considered as a set, i.e. a list of elements. For example, which elements $a_1, a_2, \ldots$, which are themselves sets (because everything is a set), would allow one to express the integer 2 as the set $\{a_1, a_2, \ldots\}$? At first glance, we might attempt to represent 2 with an arbitrary set containing two elements, say {"cow", "chicken"}. The intuition here might be that the number of elements reflects our notion of the "quantity" or "magnitude" we associate to the number 2. However, upon further reflection, such a set might not satisfy other notions we associate to the number 2.

For example we typically think of the number 2 as having a unique "ranking" or "place" in the number line. That is, the number 2 is the integer that preceeds the number 3. So we would need to not only define a set representing the number 3, but also a set representing the notion of subsequence that would align with our intuition that 3 is subsequent to 2.

In a different vein, one might also wish for our representation of the number 2 to agree with the idea that the sum of 2 and 4 is 6. The issue here is how to describe the notion of integer addition. After some thought, one might realize that the idea of adding two integers is basically that of a function, i.e. an object which eats two integer inputs and spits out an integer output.

It might seem tricky to be able to represent all the nonnegative integers, the notion of integer subsequence and the notion of integer addition by sets in a way that aligns with our intution of those objects, relations and operations. With our limited toolset, this would indeed be quite challenging, but after a few more ideas, this will become a relatively simple goal. For now, we focus on providing the necessary calculus of sets to be able to do so.

The first two relations between sets that we define pertain to when one set is a subset of another and when one set is equal to another. To motivate this, we observe that each element of $\{a, b\}$ is also an element of $\{a, b, c\}$. Our intuition dictates that the former is in some sense "less than or equal to" the latter. However, the same could not be said of the converse. Therefore, the two sets should not be considered equal.

Definition 2.1.0.3. Let A and B be sets. Then

- A is said to be a **subset of** B if for each set x, $x \in A$ implies that $x \in B$.
  (A is a subset of $B \iff \forall x(x \in A \implies x \in B)$)
- A is said to be **equal to** B if $A \subset B$ and $B \subset A$.
- A is said to be **empty** if for each set x, $x \notin A$.
  (A is empty $\iff \forall x(\neg(x \in A))$)

Note 2.1.0.4.

- When defining a term, it is conventional to use an "if", but really the relationship is an "iff".
- For convenience, we will typically write $x \notin A$ in place of $\neg(x \in A)$.

The following exercise shows us that there is at most one set that is empty.

Exercise 2.1.0.5. Let A, B be sets. If A is empty and B is empty, then $A = B$.

Proof. Suppose that A is empty and B is empty. Let x be a set.

- Since A is empty, $x \notin A$. Therefore $x \in A$ implies that $x \in B$. Since x is an arbitrary set, we have that for each set x, $x \in A$ implies that $x \in B$. Hence $A \subset B$.
- Similarly $B \subset A$

Since $A \subset B$ and $B \subset A$, we have that $A = B$. □

Even though there can be at most one set that is empty, we are not guaranteed the existence of such a set. Since its existence is quite useful and intuitively quite reasonable, we will assume such a set does in fact exist.

Axiom 2.1.0.6. There exists a set $\varnothing$, referred to as the **empty set**, such that for each set x, $x \notin \varnothing$.

Exercise 2.1.0.7. Let A be a set. Then $\varnothing \subset A$.

Proof. Let x be a set. Then $x \notin \varnothing$. Hence $x \in \varnothing$ implies that $x \in A$. Since x is an arbitrary set, we have that for each set x, $x \in \varnothing$ implies that $x \in A$. Thus $\varnothing \subset A$. □

give intuition for union, intersection, complement, show we can define union in terms of intersection and complement, call back to demorgans laws for logic

Definition 2.1.0.8. Let A, B be sets. We define intersection, union, empty set

discuss and introduce/make pictures of venn diagrams

discuss how definitions by convention use an if, but it should be treated as an iff. The if makes sense because if we say that a dog is "blarge" if it is black and weighs more than 15 kg, if we had put an iff in this definition, then the we would be saying that if a dog is blarge, then it is black and weighs more than 15 kgs, which wouldn really make sense in a definition because we have not defined previously and do not know what it is blarge, so it is a language thing. but really we are defining a set. $x \in \{\text{blarge dogs}\} \iff x \in \{\text{black dogs}\}\text{and}\{\text{dogs above 15 kg}\}$ i.e. $\{\text{blarge dogs}\} = \{\text{black dogs}\} \cap \{\text{dogs above 15 kg}\}$.

discuss how we can define natural numbers

discuss how we define $(a, b) = \{\{a\}, \{a, b\}\}$ or similar

Definition 2.1.0.9. products $A \times B$

Exercise 2.1.0.10. content...

Exercise 2.1.0.11.

2.2 Relations

keep this part here, after section on sets similar to analysis notes, so we can assume arbitrary unions, products, etc, not just unions/products/etc of two sets

2.2.1 Introduction

Definition 2.2.1.1. Let A, B be sets.

- Let $R \subset A \times B$. Then R is said to be a **relation between** A **and** B.
- We define $\mathcal{R}(A, B) := \mathcal{P}(A, B)$

Definition 2.2.1.2. Let A, B, C be sets, $R \in \mathcal{R}(A, B)$, $S \in \mathcal{R}(B, C)$. We define the **composition relation of** S **with** R, denoted $S \circ_{\mathbf{Rel}} R \in \mathcal{R}(A, C)$, by

$$S \circ_{\mathbf{Rel}} R := \{(a, c) \in A \times C : \text{there exists } b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in S\}.$$

Note 2.2.1.3. When the context is clear, we write $S \circ R$ in place of $S \circ_{\mathbf{Rel}} R$.

Exercise 2.2.1.4. Let A, B, C, D be sets, R a relation between A and B, S relation between B and C and T relation between C and D. Then $(T \circ S) \circ R = T \circ (S \circ R)$.

Proof.

- Let $(a, d) \in (T \circ S) \circ R$. Then there exists $b \in B$ such that $(a, b) \in R$ and $(b, d) \in T \circ S$. Since $(b, d) \in T \circ S$, there exists $c \in C$ such that $(b, c) \in S$ and $(c, d) \in T$. Since $(a, b) \in R$ and $(b, c) \in S$, $(a, c) \in S \circ R$. Since $(a, c) \in S \circ R$ and $(c, d) \in T$, we have that $(a, d) \in T \circ (S \circ R)$. Since $(a, d) \in (T \circ S) \circ R$ is arbitrary, we have that for each $(a, d) \in (T \circ S) \circ R$, $(a, d) \in T \circ (S \circ R)$. Hence $(T \circ S) \circ R \subset T \circ (S \circ R)$.
- Similarly, $T \circ (S \circ R) \subset (T \circ S) \circ R$.

Since $(T \circ S) \circ R \subset T \circ (S \circ R)$ and $T \circ (S \circ R) \subset (T \circ S) \circ R$, we have that $(T \circ S) \circ R = T \circ (S \circ R)$. □

Definition 2.2.1.5. Let A be a set. We define the **diagonal relation on** A, denoted $\Delta_A \in \mathcal{R}(A, A)$, by

$$\Delta_A := \{(a, a) : a \in A\}.$$

Exercise 2.2.1.6. Let A, B be sets and $R \in \mathcal{R}(A, B)$. Then

1. $R \circ \Delta_A = R$
2. $\Delta_B \circ R = R$

Proof.

1.
   - Let $(a, b) \in R \circ \Delta_A$. Then there exists $a' \in A$ such that $(a, a') \in \Delta_A$ and $(a', b) \in R$. By definition of Δ_A, $a = a'$. Therefore $(a, b) \in R$. Since $(a, b) \in R \circ \Delta_A$ is arbitrary, we have that for each $(a, b) \in R \circ \Delta_A$, $(a, b) \in R$. Hence $R \circ \Delta_A \subset R$.
   - Let $(a, b) \in R$. Then $a \in A$. Therefore $(a, a) \in \Delta_A$. Since $(a, a) \in \Delta_A$ and $(a, b) \in R$, we have that $(a, b) \in R \circ \Delta_A$. Since $(a, b) \in R$ is arbitrary, we have that for each $(a, b) \in R$, $(a, b) \in R \circ \Delta_A$. Hence $R \circ \Delta_A \subset R$.

   Since $R \circ \Delta_A \subset R$ and $R \circ \Delta_A \subset R$, we have that $R \circ \Delta_A = R$.
2. Similar to (1).

□

Exercise 2.2.1.7. Let A, B, C be sets.

1. Let $R \in \mathcal{R}(A, B)$ and $(S_\gamma)_{\gamma \in \Gamma} \subset \mathcal{R}(B, C)$. Then

(a) $\left(\bigcup_{\gamma\in\Gamma} S_\gamma\right)\circ R = \bigcup_{\gamma\in\Gamma}(S_\gamma\circ R)$,

(b) $\left(\bigcap_{\gamma\in\Gamma} S_\gamma\right)\circ R \subset \bigcap_{\gamma\in\Gamma}(S_\gamma\circ R)$

2. Let $(R_\gamma)_{\gamma\in\Gamma}\subset\mathcal{R}(A,B)$ and $S\in\mathcal{R}(B,C)$. Then

(a) $S\circ\left(\bigcup_{\gamma\in\Gamma} R_\gamma\right) = \bigcup_{\gamma\in\Gamma}(S\circ R_\gamma)$.

(b) $S\circ\left(\bigcap_{\gamma\in\Gamma} R_\gamma\right) \subset \bigcap_{\gamma\in\Gamma}(S\circ R_\gamma)$.

Proof.

1. (a) We note that

$$\begin{aligned}
(a,c)\in\left(\bigcup_{\gamma\in\Gamma} S_\gamma\right)\circ R &\iff \text{there exists } b\in B \text{ such that } (a,b)\in R \text{ and } (b,c)\in\bigcup_{\gamma\in\Gamma} S_\gamma\\
&\iff \text{there exists } b\in B \text{ and } \gamma_0\in\Gamma \text{ such that } (a,b)\in R \text{ and } (b,c)\in S_{\gamma_0}\\
&\iff \text{there exists } \gamma_0\in\Gamma \text{ such that } (a,c)\in S_{\gamma_0}\circ R\\
&\iff (a,c)\in\bigcup_{\gamma\in\Gamma}(S_\gamma\circ R).
\end{aligned}$$

Hence $\left(\bigcup_{\gamma\in\Gamma} S_\gamma\right)\circ R = \bigcup_{\gamma\in\Gamma}(S_\gamma\circ R)$.

(b) We note that

$$\begin{aligned}
(a,c)\in\left(\bigcap_{\gamma\in\Gamma} S_\gamma\right)\circ R &\iff \text{there exists } b\in B \text{ such that } (a,b)\in R \text{ and } (b,c)\in\bigcap_{\gamma\in\Gamma} S_\gamma\\
&\iff \text{there exists } b\in B \text{ such that } (a,b)\in R \text{ and for each } \gamma\in\Gamma \text{ and } (b,c)\in S_\gamma\\
&\implies \text{for each } \gamma\in\Gamma, \text{ there exists } b\in B \text{ such that } (a,b)\in R \text{ and } (b,c)\in S_\gamma\\
&\iff \text{for each } \gamma\in\Gamma,\ (a,c)\in S_\gamma\circ R\\
&\iff (a,c)\in\bigcap_{\gamma\in\Gamma}(S_\gamma\circ R).
\end{aligned}$$

Hence $\left(\bigcap_{\gamma\in\Gamma} S_\gamma\right)\circ R = \bigcap_{\gamma\in\Gamma}(S_\gamma\circ R)$.

2. (a) We note that

$$\begin{aligned}
(a,c)\in S\circ\left(\bigcup_{\gamma\in\Gamma} R_\gamma\right) &\iff \text{there exists } b\in B \text{ such that } (a,b)\in\bigcup_{\gamma\in\Gamma} R_\gamma \text{ and } (b,c)\in S\\
&\iff \text{there exists } b\in B \text{ and } \gamma_0\in\Gamma \text{ such that } (a,b)\in R_{\gamma_0} \text{ and } (b,c)\in S\\
&\iff \text{there exists } \gamma_0\in\Gamma \text{ such that } (a,c)\in S\circ R_{\gamma_0}\\
&\iff (a,c)\in\bigcup_{\gamma\in\Gamma}(S\circ R_\gamma).
\end{aligned}$$

Hence $S\circ\left(\bigcup_{\gamma\in\Gamma} R_\gamma\right) = \bigcup_{\gamma\in\Gamma}(S_\gamma\circ R)$.

(b) We note that

$$\begin{aligned}
(a,c) \in S \circ \left(\bigcap_{\gamma \in \Gamma} R_\gamma \right) &\iff \text{there exists } b \in B \text{ such that } (a,b) \in \bigcap_{\gamma \in \Gamma} R_\gamma \text{ and } (b,c) \in S \\
&\iff \text{there exists } b \in B \text{ such that for each } \gamma \in \Gamma,\ (a,b) \in R_\gamma \text{ and } (b,c) \in S \\
&\implies \text{for each } \gamma \in \Gamma, \text{ there exists } b \in B \text{ such that } (a,b) \in R_\gamma \text{ and } (b,c) \in S \\
&\iff \text{for each } \gamma \in \Gamma,\ (a,c) \in S \circ R_\gamma \\
&\iff (a,c) \in \bigcap_{\gamma \in \Gamma} (S \circ R_\gamma).
\end{aligned}$$

Hence $S \circ \left(\bigcap_{\gamma \in \Gamma} R_\gamma \right) \subset \bigcap_{\gamma \in \Gamma} (S_\gamma \circ R)$.

□

Exercise 2.2.1.8. Let A, B, C be sets.

1. Let $R_1, R_2 \in \mathcal{R}(A,B)$ and $S \in \mathcal{R}(B,C)$. If $R_1 \subset R_2$, then $S \circ R_1 \subset S \circ R_2$.
2. Let $R \in \mathcal{R}(A,B)$ and $S_1, S_2 \in \mathcal{R}(B,C)$. If $S_1 \subset S_2$, then $S_1 \circ R \subset S_2 \circ R$.
3. Let $R_1, R_2 \in \mathcal{R}(A,B)$ and $S_1, S_2 \in \mathcal{R}(B,C)$. If $R_1 \subset R_2$ and $S_1 \subset S_2$, then $S_1 \circ R_1 \subset S_2 \circ R_2$.

Proof.

1. Suppose that $R_1 \subset R_2$. Then

$$\begin{aligned}
(a,c) \in S \circ R_1 &\iff \text{there exists } b \in B \text{ such that } (a,b) \in R_1 \text{ and } (b,c) \in S \\
&\implies \text{there exists } b \in B \text{ such that } (a,b) \in R_2 \text{ and } (b,c) \in S \\
&\iff (a,c) \in S \circ R_2.
\end{aligned}$$

   Hence $S \circ R_1 \subset S \circ R_2$.

2. Suppose that $S_1 \subset S_2$. Then

$$\begin{aligned}
(a,c) \in S_1 \circ R &\iff \text{there exists } b \in B \text{ such that } (a,b) \in R \text{ and } (b,c) \in S_1 \\
&\implies \text{there exists } b \in B \text{ such that } (a,b) \in R \text{ and } (b,c) \in S_2 \\
&\iff (a,c) \in S_2 \circ R.
\end{aligned}$$

   Hence $S_2 \circ R \subset S_1 \circ R$.

3. Suppose that $R_1 \subset R_2$ and $S_1 \subset S_2$. Then the previous parts imply that

$$\begin{aligned}
S_1 \circ R_1 &\subset S_2 \circ R_1 \\
&\subset S_2 \circ R_2.
\end{aligned}$$

□

Definition 2.2.1.9. Let A, B be sets and $R \in \mathcal{R}(A,B)$. We define the **converse of** R, denoted $R^* \in \mathcal{R}(B,A)$, by

$$R^* := \{(b,a) \in B \times A : (a,b) \in R\}.$$

Exercise 2.2.1.10. Let A, B be sets and R a relation between A and B. Then $(R^*)^* = R$.

Proof.

$$\begin{aligned}
(a,b) \in (R^*)^* &\iff (b,a) \in R^* \\
&\iff (a,b) \in R.
\end{aligned}$$

Hence $(R^*)^* = R$. □

Exercise 2.2.1.11. Let A, B, C be sets, $R \in \mathcal{R}(A, B)$ and $S \in \mathcal{R}(B, C)$. Then $(S \circ R)^* = R^* \circ S^*$.

Proof. We note that

$$\begin{aligned}
(c, a) \in (S \circ R)^* &\iff (a, c) \in S \circ R \\
&\iff \text{there exists } b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in S \\
&\iff \text{there exists } b \in B \text{ such that } (c, b) \in S^* \text{ and } (b, a) \in R^* \\
&\iff (c, a) \in R^* \circ S^*.
\end{aligned}$$

Hence $(S \circ R)^* = R^* \circ S^*$. □

Exercise 2.2.1.12. Let A and B be sets and $(R_\gamma)_{\gamma \in \Gamma} \subset \mathcal{R}(A, B)$. Then

1. $\left(\bigcup_{\gamma \in \Gamma} R_\gamma \right)^* = \bigcup_{\gamma \in \Gamma} R_\gamma^*$
2. $\left(\bigcap_{\gamma \in \Gamma} R_\gamma \right)^* = \bigcap_{\gamma \in \Gamma} R_\gamma^*$

Proof.

1. We note that

$$\begin{aligned}
(b, a) \in \left(\bigcup_{\gamma \in \Gamma} R_\gamma \right)^* &\iff (a, b) \in \bigcup_{\gamma \in \Gamma} R_\gamma \\
&\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (a, b) \in R_{\gamma_0} \\
&\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (b, a) \in R_{\gamma_0}^* \\
&\iff (b, a) \in \bigcup_{\gamma \in \Gamma} R_\gamma^*.
\end{aligned}$$

   Hence $\left(\bigcup_{\gamma \in \Gamma} R_\gamma \right)^* = \bigcup_{\gamma \in \Gamma} R_\gamma^*$.

2. We note that

$$\begin{aligned}
(b, a) \in \left(\bigcap_{\gamma \in \Gamma} R_\gamma \right)^* &\iff (a, b) \in \bigcap_{\gamma \in \Gamma} R_\gamma \\
&\iff \text{for each } \gamma \in \Gamma,\ (a, b) \in R_\gamma \\
&\iff \text{for each } \gamma \in \Gamma,\ (b, a) \in R_\gamma^* \\
&\iff (b, a) \in \bigcap_{\gamma \in \Gamma} R_\gamma^*.
\end{aligned}$$

   Hence $\left(\bigcap_{\gamma \in \Gamma} R_\gamma \right)^* = \bigcap_{\gamma \in \Gamma} R_\gamma^*$.

□

2.2.2 Subrelations

Definition 2.2.2.1. Let A, B be sets, $R \in \mathcal{R}(A, B)$, $E \subset A$ and $F \subset B$. We define the **subrelation of R between E and F**, denoted $R|_{E \times F} \in \mathcal{R}(E, F)$, by $R|_{E \times F}$

maybe dont do this subsection, since relations are like morphisms, so unless you can find a categorical way to present restrictions on domain or codomain $f|_E$ or $f|^F$, just get rid of this subsection

2.2.3 Products of Relations

Definition 2.2.3.1. Let A, B, C, D be sets and $R \in \mathcal{R}(A, B), S \in \mathcal{R}(C, D)$. We define the **product relation of R and S**, denoted $R \times_{\mathbf{Rel}} S \in \mathcal{R}(A \times C, B \times D)$, by

$$R \times_{\mathbf{Rel}} S := \{((a, c), (b, d)) \in (A \times C) \times (B \times D) : (a, b) \in R \text{ and } (c, d) \in S\}.$$

Note 2.2.3.2. When the context is clear, we write $R \times S$ in place of $R \times_{\mathbf{Rel}} S$.

Exercise 2.2.3.3. Let A, B, C, D be sets.

1. Let $R \in \mathcal{R}(A, B)$ and $(S_\gamma)_{\gamma \in \Gamma} \subset \mathcal{R}(C, D)$. Then
   (a) $R \times \left(\bigcup_{\gamma \in \Gamma} S_\gamma \right) = \bigcup_{\gamma \in \Gamma} (R \times S_\gamma)$
   (b) $R \times \left(\bigcap_{\gamma \in \Gamma} S_\gamma \right) = \bigcap_{\gamma \in \Gamma} (R \times S_\gamma)$
2. Let $(R_\gamma)_{\gamma \in \Gamma} \subset\in \mathcal{R}(A, B)$ and $S \in \mathcal{R}(C, D)$. Then
   (a) $\left(\bigcup_{\gamma \in \Gamma} R_\gamma \right) \times S = \bigcup_{\gamma \in \Gamma} (R_\gamma \times S)$
   (b) $\left(\bigcap_{\gamma \in \Gamma} R_\gamma \right) \times S = \bigcap_{\gamma \in \Gamma} (R_\gamma \times S)$

Proof.

1. (a) We note that

$$\begin{aligned}
((a, c), (b, d)) \in R \times \left(\bigcup_{\gamma \in \Gamma} S_\gamma \right) &\iff (a, b) \in R \text{ and } (c, d) \in \bigcup_{\gamma \in \Gamma} S_\gamma \\
&\iff (a, b) \in R \text{ and there exists } \gamma_0 \in \Gamma \text{ such that } (c, d) \in S_{\gamma_0} \\
&\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (a, b) \in R \text{ and } (c, d) \in S_{\gamma_0} \\
&\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } ((a, c), (b, d)) \in R \times S_{\gamma_0} \\
&\iff ((a, c), (b, d)) \in \bigcup_{\gamma \in \Gamma} (R \times S_\gamma).
\end{aligned}$$

   Hence $R \times \left(\bigcup_{\gamma \in \Gamma} S_\gamma \right) = \bigcup_{\gamma \in \Gamma} (R \times S_\gamma)$.

   (b) We note that

$$\begin{aligned}
((a, c), (b, d)) \in R \times \left(\bigcap_{\gamma \in \Gamma} S_\gamma \right) &\iff (a, b) \in R \text{ and } (c, d) \in \bigcap_{\gamma \in \Gamma} S_\gamma \\
&\iff (a, b) \in R \text{ for each } \gamma \in \Gamma,\ (c, d) \in S_\gamma \\
&\iff \text{for each } \gamma \in \Gamma,\ (a, b) \in R \text{ and } (c, d) \in S_\gamma \\
&\iff \text{for each } \gamma \in \Gamma,\ ((a, c), (b, d)) \in R \times S_\gamma \\
&\iff ((a, c), (b, d)) \in \bigcap_{\gamma \in \Gamma} (R \times S_\gamma).
\end{aligned}$$

   Hence $R \times \left(\bigcap_{\gamma \in \Gamma} S_\gamma \right) = \bigcap_{\gamma \in \Gamma} (R \times S_\gamma)$.

2. (a) We note that

$$\begin{aligned}((a,c),(b,d)) \in \left(\bigcup_{\gamma\in\Gamma} R_\gamma\right) \times S &\iff (a,b) \in \left(\bigcup_{\gamma\in\Gamma} R_\gamma\right) \text{ and } (c,d) \in S \\ &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (a,b) \in R_{\gamma_0} \text{ and } (c,d) \in S \\ &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } ((a,c),(b,d)) \in R_{\gamma_0} \times S \\ &\iff ((a,c),(b,d)) \in \bigcup_{\gamma\in\Gamma} (R_\gamma \times S)\end{aligned}$$

Hence $\left(\bigcup_{\gamma\in\Gamma} R_\gamma\right) \times S = \bigcup_{\gamma\in\Gamma} (R_\gamma \times S)$.

(b) We note that

$$\begin{aligned}((a,c),(b,d)) \in \left(\bigcap_{\gamma\in\Gamma} R_\gamma\right) \times S &\iff (a,b) \in \left(\bigcap_{\gamma\in\Gamma} R_\gamma\right) \text{ and } (c,d) \in S \\ &\iff \text{for each } \gamma \in \Gamma,\ (a,b) \in R_\gamma \text{ and } (c,d) \in S \\ &\iff \text{for each } \gamma \in \Gamma,\ ((a,c),(b,d)) \in R_\gamma \times S \\ &\iff ((a,c),(b,d)) \in \bigcap_{\gamma\in\Gamma} (R_\gamma \times S)\end{aligned}$$

Hence $\left(\bigcap_{\gamma\in\Gamma} R_\gamma\right) \times S = \bigcap_{\gamma\in\Gamma} (R_\gamma \times S)$.

$\square$

2.3 Equivalence Relations

Note 2.3.0.1. Let X be a set and $\nu \in \mathcal{R}^2(X)$. When the context is clear, we may write $a\nu b$ in place of $(a, b) \in \nu$.

Definition 2.3.0.2. Let X be a set. Suppose that $X \neq \varnothing$.

- Let $\theta \in \mathcal{R}^2(X)$. Then θ is said to be
  - **reflexive** if for each $a \in X$, $(a, a) \in \theta$
  - **symmetric** if for each $a, b \in X$, $(a, b) \in \theta$ implies that $(b, a) \in \theta$
  - **transitive** if for each $a, b, c \in X$, $(a, b) \in \theta$ and $(b, c) \in \theta$ implies that $(a, c) \in \theta$
  - an **equivalence relation on** X if θ is reflexive, symmetric and transitive
- We define $\mathrm{Equiv}(X) := \{\theta \in \mathcal{R}^2(X) : \theta \text{ is an equivalence relation on } X\}$

Exercise 2.3.0.3. Let X be a set and $\theta \in \mathcal{R}^2(X)$. Then

1. θ is reflexive iff $\Delta_X \subset \theta$
2. θ is symmetric iff $\theta^* \subset \theta$
3. θ is transitive iff $\theta \circ \theta \subset \theta$

Proof.

1.
   - $(\implies)$:
     Suppose that θ is reflexive. Let $(a, a) \in \Delta_X$. Then $a \in X$. Since θ is reflexive, $(a, a) \in \theta$. Since $(a, a) \in \Delta_X$ is arbitrary, we have that for each $(a, a) \in \Delta_X$, $(a, a) \in \theta$. Hence $\Delta_X \subset \theta$.
   - $(\impliedby)$:
     Suppose that $\Delta_X \subset \theta$. Let $a \in X$. By definition and assumption,
     $$\begin{aligned}(a, a) &\in \Delta_X \\ &\subset \theta.\end{aligned}$$
     Since $a \in X$ is arbitrary, we have that for each $a \in X$, $(a, a) \in \theta$. Hence θ is reflexive.
2.
   - $(\implies)$:
     Suppose that θ is symmetric. Let $(b, a) \in \theta^*$. By definition, $(a, b) \in \theta$. Since θ is symmetric, $(b, a) \in \theta$. Since $(b, a) \in \theta^*$ is arbitrary, we have that for each $(b, a) \in \theta^*$, $(b, a) \in \theta$. Hence $\theta^* \subset \theta$.
   - $(\impliedby)$:
     Suppose that $\theta^* \subset \theta$. Let $a, b \in X$. Suppose that $(a, b) \in \theta$. By definition and assumption,
     $$\begin{aligned}(b, a) &\in \theta^* \\ &\subset \theta.\end{aligned}$$
     Since $a, b \in X$ with $(a, b) \in \theta$ are arbitrary, we have that for each $a, b \in X$, $(a, b) \in \theta$ implies that $(b, a) \in \theta$. Hence θ is symmetric.
3.
   - $(\implies)$:
     Suppose that θ is transitive. Let $(a, c) \in \theta \circ \theta$. By definition, there exists $b \in X$ such that $(a, b) \in \theta$ and $(b, c) \in \theta$. Since θ is transitive, $(a, c) \in \theta$. Since $(a, c) \in \theta \circ \theta$ is arbitrary, we have that for each $(a, c) \in \theta \circ \theta$, $(a, c) \in \theta$. Hence $\theta \circ \theta \subset \theta$.
   - $(\impliedby)$:
     Suppose that $\theta \circ \theta \subset \theta$. Let $a, b, c \in X$. Suppose that $(a, b) \in \theta$ and $(b, c) \in \theta$. By definition and assumption,
     $$\begin{aligned}(a, c) &\in \theta \circ \theta \\ &\subset \theta.\end{aligned}$$
     Since $a, b, c \in X$ with $(a, b) \in \theta$ and $(b, c) \in \theta$ are arbitrary, we have that for each $a, b \in X$, $(a, b) \in \theta$ and $(b, c) \in \theta$ implies that $(a, c) \in \theta$. Hence θ is transitive.

□

Definition 2.3.0.4. Let X be a set. Suppose that $X \neq \varnothing$. We define the **universal relation on** X, denoted $\nabla_X \in \mathcal{R}^2(X)$, by $\nabla_X := X^2$.

Exercise 2.3.0.5. Let X be a set. Suppose that $X \neq \varnothing$. Then $\nabla_X \in \mathrm{Equiv}(X)$.

Proof. Let $a, b, c \in X$.

1. Since $a \in X$, $(a, a) \in \nabla_X$.
2. Suppose that $(a, b) \in \nabla_X$. Since $a, b \in X$, $(b, a) \in X^2$. Therefore $(a, b) \in \nabla_X$ implies that $(b, a) \in \nabla_X$.
3. Suppose that $(a, b), (b, c) \in \nabla_X$. Since $a, c \in X$, $(a, c) \in X^2$. Therefore $(a, b), (b, c) \in X^2$ implies that $(a, c) \in \nabla_X$.

□

Exercise 2.3.0.6. Let X be a set. Suppose that $X \neq \varnothing$. Then $\Delta_X \in \mathrm{Equiv}(X)$.

Proof. Let $a, b, c \in X$.

1. Since $a \in X$, $(a, a) \in \Delta_X$.
2. Suppose that $(a, b) \in \Delta_X$. Then $a = b$ and
$$\begin{aligned}(b, a) &= (a, a)\\ &\in \Delta_X.\end{aligned}$$
Therefore $(a, b) \in \Delta_X$ implies that $(b, a) \in \Delta_X$.
3. Suppose that $(a, b), (b, c) \in \Delta_X$. Then $a = b$ and $b = c$. Hence $a = c$ and
$$\begin{aligned}(a, c) &= (a, a)\\ &\in \Delta_X.\end{aligned}$$
Therefore $(a, b), (b, c) \in \Delta_X$ implies that $(a, c) \in \Delta_X$.

□

Exercise 2.3.0.7. Let X be a set. Suppose that $X \neq \varnothing$. Then for each $(\theta_\alpha)_{\alpha \in A} \subset \mathrm{Equiv}(X)$, $\bigcap_{\alpha \in A} \theta_\alpha \in \mathrm{Equiv}(X)$.

Proof. Let $(\theta_\alpha)_{\alpha \in A} \subset \mathrm{Equiv}(X)$. Set $\theta := \bigcap_{\alpha \in A} \theta_\alpha$. Let $a, b, c \in X$.

1. Let $\alpha \in A$. Since $\theta_\alpha \in \mathrm{Equiv}(X)$, $(a, a) \in \theta_\alpha$. Since $\alpha \in A$ is arbitrary, we have that for each $\alpha \in A$, $(a, a) \in \theta_\alpha$. Hence $(a, a) \in \theta$.
2. Suppose that $(a, b) \in \theta$. Let $\alpha \in A$. Then $(a, b) \in \theta_\alpha$. Since $\theta_\alpha \in \mathrm{Equiv}(X)$, we have that $(b, a) \in \theta_{a\langle}$. Since $\alpha \in A$ is arbitrary, we have that for each $\alpha \in A$, $(b, a) \in \theta_\alpha$. Hence $(b, a) \in \theta$. Therefore $(a, b) \in \theta$ implies that $(b, a) \in \theta$.
3. Suppose that $(a, b) \in \theta$ and $(b, c) \in \theta$. Let $\alpha \in A$. Then $(a, b), (b, c) \in \theta_\alpha$. Since $\theta_\alpha \in \mathrm{Equiv}(X)$, $(a, c) \in \theta_\alpha$. Since $\alpha \in A$ is arbitrary, we have that for each $\alpha \in A$, $(a, c) \in \theta_\alpha$. Hence $(a, c) \in \theta$. Therefore $(a, b) \in \theta$ and $(b, c) \in \theta$ implies that $(a, c) \in \theta$.

Hence $\theta \in \mathrm{Equiv}(X)$. □

Definition 2.3.0.8. Let X be a set. Suppose that $X \neq \varnothing$. Let $\nu \in \mathcal{R}^2(X)$.

- We define the **equivalence relations containing** ν, denoted $\mathcal{E}(\nu)$, by $\mathcal{E}(\nu) := \{\theta \in \mathrm{Equiv}(X) : \nu \subset \theta\}$.
- We define the **equivalence relation on** X **generated by** ν, denoted $\epsilon(\nu) \in \mathcal{R}^2(X)$ by $\epsilon(\nu) := \bigcap_{\theta \in \mathcal{E}(\nu)} \theta$.

Note 2.3.0.9.

- A previous ex implies that $\nabla_X \in \mathcal{E}(\nu)$.
- A previous ex (about intersections of equiv relats) implies that $\epsilon(\nu) \in \mathcal{E}(\nu)$.

Exercise 2.3.0.10. Let X be a set. Suppose that $X \neq \varnothing$. Let $\nu \in \mathcal{R}^2(X)$. Define $(\nu_n)_{n\in\mathbb{N}_0} \subset \mathcal{R}^2(X)$ by

$$\nu_n := \begin{cases} \Delta_X \cup \nu \cup \nu^*, & n = 0 \\ \nu_{n-1} \circ \nu_{n-1}, & n \geq 1. \end{cases}$$

Then

1. (a) for each $n \in \mathbb{N}$, ν_n is reflexive
   (b) for each $n \in \mathbb{N}$, ν_n is symmetric
   (c) for each $n \in \mathbb{N}_0$, $\nu_n \subset \nu_{n+1}$
2. $\epsilon(\nu) = \bigcup_{n\in\mathbb{N}_0} \nu_n$.

Proof.

1. (a)
   - **Base Case:**
     Since
     $$\begin{aligned} \Delta_X &\subset \Delta_X \cup \nu \cup \nu^* \\ &= \nu_0, \end{aligned}$$
     a previous ex implies that ν_0 is reflexive.
   - **Induction Step:**
     Let $n \in \mathbb{N}_0$. Suppose that ν_n is reflexive. Then that same prev ex implies that $\Delta_X \subset \nu_n$. two prev ex in section on relations about composition with identity and subsets then imply that
     $$\begin{aligned} \Delta_X &= \Delta_X \circ \Delta_X \\ &\subset \nu_n \circ \nu_n \\ &= \nu_{n+1}. \end{aligned}$$
     Another application of that prev ex implies that ν_{n+1} is reflexive.

   By induction, for each $n \in \mathbb{N}_0$, ν_n is reflexive.

   (b)
   - **Base Case:**
     a previous ex about converse relations and unions, double converse and since $\Delta_X \in \mathrm{Equiv}(X)$ and an ex about symmetric relations implies that
     $$\begin{aligned} \nu_0^* &= (\Delta_X \cup \nu \cup \nu^*)^* \\ &= \Delta_X^* \cup \nu^* \cup (\nu^*)^* \\ &= \Delta_X^* \cup \nu^* \cup \nu \\ &\subset \Delta_X \cup \nu^* \cup \nu \\ &= \Delta_X \cup \nu \cup \nu^* \\ &= \nu_0. \end{aligned}$$
     that same previous ex about symmetric relations implies that ν_0 is symmetric.
   - **Induction Step:**
     Let $n \in \mathbb{N}_0$. Suppose that ν_n is symmetric. Then that same prev ex about symmetric and converse relations implies that $\nu_n^* \subset \nu_n$. Therefore prev exs imply
     $$\begin{aligned} \nu_{n+1}^* &= (\nu_n \circ \nu_n)^* \\ &= \nu_n^* \circ \nu_n^* \\ &\subset \nu_n \circ \nu_n. \end{aligned} \qquad = \nu_{n+1}.$$

Another application of that prev ex implies that ν_{n+1} is symmetric.

By induction, for each $n \in \mathbb{N}_0$, ν_n is symmetric.

(c) Let $n \in \mathbb{N}_0$. Since ν_n is reflexive, prev ex implies that $\Delta_X \subset \nu_n$. Therefore prev ex implies

$$\begin{aligned}\nu_n &= \Delta_X \circ \nu_n \\ &\subset \nu_n \circ \nu_n \\ &= \nu_{n+1}.\end{aligned}$$

Since $n \in \mathbb{N}_0$ is arbitrary, we have that for each $n \in \mathbb{N}_0$, $\nu_n \subset \nu_{n+1}$.

2. Set $\theta_0 := \bigcup_{n \in \mathbb{N}_0} \nu_n$.

- – **Base Case:**
  Since $\epsilon(\nu) \in \mathcal{E}(\nu)$, we have that $\epsilon(\nu) \in \mathrm{Equiv}(X)$ and $\nu \subset \epsilon(\nu)$. Since $\epsilon(\nu) \in \mathrm{Equiv}(X)$, we have that $\epsilon(\nu)$ is reflexive and symmetric. A ref ex then implies that $\Delta_X \subset \epsilon(\nu)$ and $\epsilon(\nu)^* \subset \epsilon(\nu)$. Therefore a prev ex about converse relations and subsets

$$\begin{aligned}\nu_0 &= \Delta_X \cup \nu \cup \nu^* \\ &\subset \epsilon(\nu) \cup \epsilon(\nu) \cup \epsilon(\nu)^* \\ &\subset \epsilon(\nu) \cup \epsilon(\nu) \cup \epsilon(\nu) \\ &= \epsilon(\nu).\end{aligned}$$

  – **Induction Step:**
  Let $n \in \mathbb{N}_0$. Suppose that $\nu_n \subset \epsilon(\nu)$. Then ref prev ex

$$\begin{aligned}\nu_{n+1} &= \nu_n \circ \nu_n \\ &\subset \epsilon(\nu) \circ \epsilon(\nu) \\ &= \epsilon(\nu).\end{aligned}$$

  By induction, for each $n \in \mathbb{N}_0$, $\nu_n \subset \epsilon(\nu)$. Therefore

$$\begin{aligned}\theta_0 &= \bigcup_{n \in \mathbb{N}_0} \nu_n \\ &\subset \epsilon(\nu).\end{aligned}$$

- – Since ν_0 is reflexive, we have that

$$\begin{aligned}\Delta_X &\subset \nu_0 \\ &\subset \bigcup_{n \in \mathbb{N}_0} \nu_n \\ &= \theta_0.\end{aligned}$$

  Hence θ_0 is reflexive.

  – Since for each $n \in \mathbb{N}_0$, ν_n is symmetric, we have that that ref exs for each $n \in \mathbb{N}_0$, $\nu_n^* \subset \nu_n$. Therefore

$$\begin{aligned}\theta_0^* &= (\bigcup_{n \in \mathbb{N}_0} \nu_n)^* \\ &= \bigcup_{n \in \mathbb{N}_0} \nu_n^* \\ &\subset \bigcup_{n \in \mathbb{N}_0} \nu_n \\ &= \theta_0.\end{aligned}$$

  Hence θ_0 is symmetric.

– Let $a, b, c \in X$. Suppose that $(a, b) \in \theta_0$ and $(b, c) \in \theta_0$. Then there exist $n_1, n_2 \in \mathbb{N}_0$ such that $(a, b) \in \nu_{n_1}$ and $(b, c) \in \nu_{n_2}$. Set $n_0 := \max\{n_1, n_2\}$. Since for each $n \in \mathbb{N}$, $\nu_n \subset \nu_{n+1}$, we have that

$$\begin{aligned}(a, b) &\in \nu_{n_1} \\ &\subset \nu_{n_0}.\end{aligned}$$

and

$$\begin{aligned}(b, c) &\in \nu_{n_2} \\ &\subset \nu_{n_0}.\end{aligned}$$

Since $(a, b), (b, c) \in \nu_{n_0}$, we have that

$$\begin{aligned}(a, c) &\in \nu_{n_0} \circ \nu_{n_0} \\ &= \nu_{n_0+1} \\ &\subset \bigcup_{n \in \mathbb{N}_0} \nu_n \\ &= \theta_0.\end{aligned}$$

Since $a, b, c \in X$ with $(a, b) \in \theta_0$ and $(b, c) \in \theta_0$ are arbitrary, we have that for each $a, b, c \in X$, $(a, b) \in \theta_0$ and $(b, c) \in \theta_0$ implies that $(a, c) \in \theta_0$. Hence θ_0 is transitive.

Since θ_0 is reflexive, symmetric and transitive, $\theta_0 \in \mathrm{Equiv}(X)$. Since

$$\begin{aligned}\nu &\subset \Delta_X \cup \nu \cup \nu^* \\ &= \nu_0 \\ &\subset \bigcup_{n \in \mathbb{N}_0} \nu_n \\ &= \theta_0,\end{aligned}$$

we have that $\theta_0 \in \mathcal{E}(\nu)$. Therefore

$$\begin{aligned}\epsilon(\nu) &= \bigcap_{\theta \in \mathcal{E}(\nu)} \theta \\ &\subset \theta_0.\end{aligned}$$

Since $\theta_0 \subset \epsilon(\nu)$ and $\epsilon(\nu) \subset \theta_0$, we have that $\theta_0 = \epsilon(\nu)$.

□

2.4 Products

Chapter 3

Operations and Relations

3.0.1 Introduction

Definition 3.0.1.1. Let X be a set.

- Let $k \in \mathbb{N}_0$.
  - Let $R \in \mathcal{P}(X^k)$ and $f \in X^{X^k}$. Then
    - R is said to be a **k-ary relation on** X
    - f is said to be a **k-ary operation on** X
  - We define
    - the **set of k-ary relations on** X, denoted $\mathcal{R}^k(X)$, by $\mathcal{R}^k(X) := \mathcal{P}(X^k)$
    - the **set of k-ary operations on** X, denoted $\mathcal{F}^k(X)$, by $\mathcal{F}^k(X) := X^{X^k}$.
- We define the
  - **set of finitary relations on** X, denoted $\mathcal{R}^*(X)$, by
  $$\mathcal{R}^*(X) := \bigcup_{k \in \mathbb{N}_0} \mathcal{R}^k(X)$$
  - **set of finitary operations on** X, denoted $\mathcal{F}^*(X)$, by
  $$\mathcal{F}^*(X) := \bigcup_{k \in \mathbb{N}_0} \mathcal{F}^k(X)$$

  set of finitary relations on X, denoted $\mathcal{R}^*(X)$, by
  $$\mathcal{R}^*(X) := \bigcup_{k \in \mathbb{N}_0} \mathcal{R}^k(X)$$
- **arity map**, denoted $\operatorname{arity} : \mathcal{R}^*(X) \to \mathbb{N}_0$, by
  $$\operatorname{arity} R := k, \quad f \in \mathcal{R}^k(X)$$
- Let X be a set, $\mathcal{R} \subset \mathcal{R}^*(X)$ and $k \in \mathbb{N}_0$. We define the **k-ary members of** $\mathcal{R}$, denoted $\mathcal{R}_k$, by
  $$\mathcal{R}_k := \mathcal{R} \cap \mathcal{R}^k(X)$$
- Let X be a set and $k \in \mathbb{N}_0$. We define the **set of k-ary operations on** X, denoted $\mathcal{F}^k(X)$, by $\mathcal{F}^k(X) := X^{(X^k)}$. We define the **set of finitary operations on** X, denoted $\mathcal{F}^*(X)$, by
  $$\mathcal{F}^*(X) := \bigcup_{k \in \mathbb{N}_0} \mathcal{F}^k(X)$$

- Let X be a set. We define the **operation arity map**, denoted $\operatorname{arity} : \mathcal{F}^*(X) \to \mathbb{N}_0$, by

$$\operatorname{arity} f := k, \quad f \in \mathcal{F}^k(X)$$

- Let X be a set, $\mathcal{F} \subset \mathcal{F}^*(X)$ and $k \in \mathbb{N}_0$. We define the **k-ary members of $\mathcal{F}$**, denoted $\mathcal{F}_k$, by

$$\mathcal{F}_k := \mathcal{F} \cap \mathcal{F}^k(X)$$

Exercise 3.0.1.2. Let X be a set. Then $\mathcal{R}^0(X) = \{\varnothing, \{\varnothing\}\}$.

Proof. Since $X^0 = \{\varnothing\}$, we have that

$$\begin{aligned} \mathcal{R}^0(X) &= \mathcal{P}(X^0) \\ &= \{\varnothing, \{\varnothing\}\}. \end{aligned}$$

□

Definition 3.0.1.3. Let X be a set, $\mathcal{F} \subset \mathcal{F}^*(X)$ and $C \subset X$. Then C is said to be **$\mathcal{F}$-closed** if for each $k \in \mathbb{N}_0$, $f \in \mathcal{F}_k$ and $a \in C^k$, $f(a) \in C$.

Exercise 3.0.1.4. Let X be a set, $\mathcal{F} \subset \mathcal{F}^*(X)$ and $\mathcal{C} \subset \mathcal{P}(X)$. If for each $C \in \mathcal{C}$, C is $\mathcal{F}$-closed, then $\bigcap_{C \in \mathcal{C}} C$ is $\mathcal{F}$-closed

need special case where $k = 0$? maybe trivially true?

Proof. Suppose that for each $C \in \mathcal{C}$, C is $\mathcal{F}$-closed. Let $k \in \mathbb{N}_0$, $f \in \mathcal{F}_k$, $a_1, \ldots, a_k \in \bigcap_{C \in \mathcal{C}} C$ and $C_0 \in \mathcal{C}$. Since $C_0 \in \mathcal{C}$, we have that

$$\begin{aligned} a_1, \ldots, a_k &\in \bigcap_{C \in \mathcal{C}} C \\ &\subset C_0 \end{aligned}$$

Since C_0 is $\mathcal{F}$-closed, we have that $f(a_1, \ldots, a_k) \in C_0$. Since $C_0 \in \mathcal{C}$ is arbitrary, we have that for each $C \in \mathcal{C}$, $f(a_1, \ldots, a_k) \in C$. Hence $f(a_1, \ldots, a_k) \in \bigcap_{C \in \mathcal{C}} C$. Since $k \in \mathbb{N}_0$ and $a_1, \ldots, a_k \in \bigcap_{C \in \mathcal{C}} C$ are arbitrary, we have that $\bigcap_{C \in \mathcal{C}} C$ is $\mathcal{F}$-closed. □

Chapter 4

Propositional Logic

Definition 4.0.0.1. Let $\mathcal{A}$ be a set, $\mathcal{V} \subset \mathcal{A}$, $\mathcal{F} \subset \mathcal{A}^*$, $\mathcal{C} \subset \bigcup_{k \in \mathbb{N}_0} \mathcal{F}^{(\mathcal{F}^k)}$. Then $(\mathcal{A}, \mathcal{V}, \mathcal{A})$ is said to be a **propositional calculus** if

1.
   - $\mathcal{V} \subset \mathcal{F}$
   - for each $k \in \mathbb{N}_0$, $p_1, \ldots, p_k \in \mathcal{F}$, and $f \in \mathcal{C}_k$, $f(p_1, \ldots, p_k) \in \mathcal{F}$
   - for each $p \in \mathcal{F}$, there exists some tree $f \in \mathcal{T}^k(\mathcal{F})$ such that $\in \mathcal{F}^k$

define the **alphabet** $\mathcal{A}$ define the **variables** $\mathcal{V}$ define the **formulas** $\mathcal{F}$ define the **connectives** $\mathcal{C}$ the **formulas of** $\mathcal{L}$**,**

Exercise 4.0.0.2.

Part I

Model Theory

4.1 ToDo

- ask chase about relation of first order logic with countable quanitification and probability with

$$\{\text{statements}\} \rightleftarrows \sigma\text{-algebra}$$

with $p \mapsto A_p$ and

$$\{\text{truth valuations}\} \rightleftarrows \{\text{probabilities}\}$$

with $v \mapsto P_v$ where the following structure is preserved:

  - $P_v(A_p) = v(p)$,
  - $P_v\left(\bigcup_{n\in\mathbb{N}} A_{p(n)}\right) = v(\exists n p(n)) = \bigvee_{n\in\mathbb{N}} v(p(n))$,
  - $P_v(A_p^c) = v(\neg p) = v(p)^*$.

  what do the borel cantelli lemma and kolmogorov 0-1 law tell us about the sentences in the language? Can we translate the notion of indpendence in probability to a notion of independence in logic using the kolmogorov 0-1 law? what about disintegration of measure? Can we frame statistics in terms of model theory? Like for a given structure $\mathcal{L}$ we have probability $P_{\mathcal{L}}$ and instead of foumlas p, we have the interpreation of measurable sets $p^{\mathcal{L}} = A_p \in \mathcal{A}$, (what about functions and constants?) except our structure is an extension of the usual model theory in where instead of the usual satisfiablity/unsatisfiablility of a formula we have $\mathcal{L} \models, \not\models p$, corresponds to $P_{\mathcal{L}}(p^{\mathcal{L}}) = 1$, $P_{\mathcal{L}}(p^{\mathcal{L}}) = 0$. Also can we explain maximum likelihood estimation as picking an optimal structure for our interpretation? can we backwards translate things like minimizing information loss (finding mle) as choosing a best structure for a given interpretation?

- try to cast project about group actions and probabilities in model theoretic language using the metric structures. The notes over metric structures talk about probability spaces as metric structure models. Can we do the same for when we care about a group action $G \times X \to X$ with the extra data of a probability μ on $(X, \mathcal{B}(X))$? We would need a metric. For general prob spaces $(X, \mathcal{A}, \mu)$, we can induce the associated probability space $(X, \hat{\mathcal{A}}, \hat{\mu})$ where $\hat{\mathcal{A}} = \mathcal{A}/\sim$ with $A \sim B$ iff $\mu(A \triangle B) = 0$ and $\hat{\pi_\sim(A)} := \mu(A)$. Then we define a metric on $\hat{\mathcal{A}}$ by $d(\pi_\sim(A), \pi_\sim(B)) = d(A, B) = \mu(A \triangle B)$. If we add a group, we might want to make a metric analogous to the case in which a group acts isometrically on a metric space, i.e. $d_{X/G}(x, y) = \inf_{g\in G} d(g \cdot x, y)$.

  We note that for any group action, since $(gA)^c = g(A^c)$, $gA \cap gB = g(A \cap B)$ and $gA \cup gB = g(A \cup B)$. Therefore

$$\begin{aligned}
(gA)\triangle(g \cdot B) &= [(gA) \cap (gB)^c] \cup [(gB) \cap (gA)^c] \\
&= [(gA) \cap (gB^c)] \cup [(gB) \cap (gA^c)] \\
&= [g(A \cap B^c) \cup g(B \cap A^c)] \\
&= g[(A \cap B^c) \cup (B \cap A^c)] \\
&= g(A \triangle B).
\end{aligned}$$

  If we assume that G preserves sets of measure zero (very mild assumption), so that for each $A \in \mathcal{A}$, $\mu(A) = 0$ implies that $\mu(gA) = 0$, then we have that $\mu(gA) = 0$ implies that $\mu(A) = \mu(g^{-1}gA) = 0$. Hence G preserves sets of measure zero implies that $\mu(gA) = 0$ iff $\mu(A) = 0$. Then

$$\begin{aligned}
A \sim B &\iff \mu(A \triangle B) = 0 \\
&\iff \mu[g(A \triangle B)] = 0 \\
&\iff \mu[(gA)\triangle(gB)] = 0 \\
&\iff gA \sim gB.
\end{aligned}$$

  Hence for each $g \in G$, $A \mapsto gA$ is $\sim$-invariant. Thus we can define $G \times \hat{\mathcal{A}} \to \hat{\mathcal{A}}$ by $g\pi_\sim(A) := \pi_\sim(gA)$ (uniqueness of $\bar{f}$ from quotient space section in analysis notes).

If we assume additionally that G preserves measures (like if $G \times X \to X$ acts isometrically, or is a hamiltonian flow (mor generally some ergodic system?)), then we have that

$$\begin{aligned}
\hat{d}(g\hat{A}, g\hat{B}) &= \hat{d}(g\pi_\sim(A), g\pi_\sim(B)) \\
&= \hat{d}(\pi_\sim(gA), \pi_\sim(gB)) \\
&= d(gA, gB) \\
&= \mu[(gA)\triangle(gB)] \\
&= \mu[g(A\triangle B)] \\
&= \mu(A\triangle B) \\
&= d(A, B) \\
&= \hat{d}(\hat{A}, \hat{B})
\end{aligned}$$

can shorten this by separating out part showing if measure preserved, then $d(gA, gB) = d(A, B)$.

Then $G \times \hat{\mathcal{A}} \to \hat{\mathcal{A}}$ acts isometrically, and if G is compact, maybe other requirements, then we can get a metric $\hat{d}_{\hat{A}/G} : \hat{A}/G \times \hat{A}/G \to [0, \infty)$ by $\hat{d}_{\hat{A}/G}(\pi_{\hat{A}/G}(\hat{A}, \pi_{\hat{A}/G}(\hat{B}))) = \inf_{g \in G} \hat{d}(g\hat{A}, \hat{B}) = \inf_{g \in G} d(gA, B)$.

would be quicker to work with pseudometric space $(\mathcal{A}, d_0)$ instead of $(\hat{\mathcal{A}}, d)$ and show in alaysis notes that that if d is a pseudometric and G acts isometrically and is compact etc, then $d_{X/G}$ is pseudometric.

4.2 Languages

Definition 4.2.0.1. Let X be a set. We define the **words over** X, denoted X^*, by

$$X^* := \bigcup_{k \in \mathbb{N}_0} X^k$$

Definition 4.2.0.2. Let X be a set.

Definition 4.2.0.3. Let $\mathcal{F}$, $\mathcal{R}$, $\mathcal{C}$ be sets and $(n_f)_{f \in \mathcal{F}}, (n_R)_{R \in \mathcal{R}} \subset \mathbb{N}$. Set $\mathcal{L} := (\mathcal{F}, \mathcal{R}, \mathcal{C}, (n_f)_{f \in \mathcal{F}}, (n_R)_{R \in \mathcal{R}})$. Then $\mathcal{L}$ is said to be a **language**. We define the

- **function symbols of** $\mathcal{L}$, denoted $\mathrm{Fun}(\mathcal{L})$, by $\mathrm{Fun}(\mathcal{L}) := \mathcal{F}$,
- **relation symbols of** $\mathcal{L}$, denoted $\mathbf{Rel}(\mathcal{L})$, by $\mathbf{Rel}(\mathcal{L}) := \mathcal{R}$,
- **constant symbols of** $\mathcal{L}$, denoted $\mathrm{Cons}(\mathcal{L})$, by $\mathrm{Cons}(\mathcal{L}) := \mathcal{C}$.

For each $f \in \mathcal{F}$ and $R \in \mathcal{R}$, we define the **arity** of f and R, denoted $\mathrm{arity}(f)$ and $\mathrm{arity}(R)$ respectively, by $\mathrm{arity}(f) := n_f$ and $\mathrm{arity}(R) := n_R$ respectively.

Note 4.2.0.4. make comment about why we distinguish between relations, functions and constants when we usually think of functions as a type of relation and constants as 0-ary functions. Seems to be a matter of convenience

Definition 4.2.0.5. Let $\mathcal{L}$ be a language, M a set, $\phi_\mathcal{F} : \mathrm{Fun}(\mathcal{L}) \to \mathcal{F}^*(M)$, $\phi_\mathcal{R} : \mathbf{Rel}(\mathcal{L}) \to \mathcal{R}^*(M)$ and $\phi_\mathcal{C} : \mathrm{Cons}(\mathcal{L}) \to M$. Set $\mathcal{M} := (M, \phi_\mathcal{F}, \phi_\mathcal{R}, \phi_\mathcal{C})$. Then $\mathcal{M}$ is said to be an **$\mathcal{L}$-structure on** M if

1. for each $f \in \mathrm{Fun}(\mathcal{L})$, $\phi_\mathcal{F}(f) \in \mathcal{F}^{n_f}(M)$,
2. for each $R \in \mathbf{Rel}(\mathcal{L})$, $\phi_\mathcal{R}(R) \in \mathcal{R}^{n_R}(M)$.

Let $f \in \mathrm{Fun}(\mathcal{L})$, $R \in \mathbf{Rel}(\mathcal{L})$ and $c \in \mathrm{Cons}(\mathcal{L})$. Then $\phi_\mathcal{F}(f)$, $\phi_\mathcal{R}(R)$ and $\phi_\mathcal{C}(c)$ are said to be **interpretations** of f, R and c in M respectively.

Chapter 5

Idea To do

idea

give brief discussion on how to make logic from a finite set theory (arbitrarily many symbols can be made from one symbol x and another symbol | by x, $x|$, $x||$ and so on, so we just need a large enough finite set for any formula with a finite number of symbols). In particular, you can define products, relations, functions, etc using finite sets and can then define logic in terms of these finite sets with a finite (but as large as you need) alphabet as a set, connectives as a set, etc.

Then talk about how once you have this finite set based logic, you can construct the usual set theory from this logic and do regular math with infinte sets using the usual axioms of set theory .

One thing that bugs me though is the idea of having finitely many "things". Assuming the set theory is true, we should be able to build a physics model of $\mathbb{R}/L\mathbb{Z}$ whose "length" L is in competetion with the number of particles N (assuming each particle is its own antiparticle like photons) where the total energy of the system is tied up in some combination of L and N. In a quantum scenario, he total system is described by the wave function of the space's/particle's field on some ambient space which is described by a lagrangian of fields so that the N and L can take on finitely many values described probabilistically and which are related via some uncertainty.

The idea being that the system oscillates between a large L and low N and small L and large N as the energy transeferes between space and particle number. t

So in this model, we can perhaps in some specific system states, momentarily define finite set theory and therefore logic in terms these ephemeral, momentarily well defined positions of particles, but the system will then evolve into a state where the finite set theory deosn't make sense. For instance in a computer, 0 or 1 only make sense when you can distinguish voltage levels, but if the threshold for identifiability is no longer achieved, then your computer will not be able to work (because the power is off or some other issue). So in this physical model, the ability to even define a finite set theory based on "things" is not possible, at least "all the time". Does mean anything for the set theory as we use it? We base basically all our eggs in a basket of assuming that we can talk about distinct finite things, but is that really true? Is there some way to salvage this in the physical model where maybe identifying things based on particle number and measurably sufficiently distinct location could be be in competition with another set of "things" such that when L is small, N is high and the above definition suffices, but when L is high and N is low, there exist another set of things we could use. But then is it possible for a state to not have enough distinguishability in either case simultaneously (at least for a given measurement precision), this seems to be the case even with regular computers. If you have very limited voltage, then even classically you can only have very small sets of things. So in some sense it seems like to have a sufficiently rich theory, you need sufficient measuremnt precision and distinguishability.

It seems like even if you couldnt recreate a finite set theory, you might be able to recreate another math structure like a group. but idk, because to do so, you ould need to be able to work with finite strings right? so basically finite sets?

Another question is if you can cook up a physical math model using eg quantum mechanical of a universe (i.e. system) such that no matter what allowable operations (unitary transformations and observables) which would allow you to recreate a given math structure like a group. For instance in our universe, we could (at least temporarily) apply the function "forget everything but my watch and wait one hour in my frame of reference" when the minute hand is "aligned" with a numeral so that the output of the operation would be the watch with the minute hand on the subsequent numeral. We could apply this over and over (at least for a little while) to recreate a theory of the cyclic group of order 12. We could cook up a much simpler model using harmonic oscillators to recreate a group as well.

The question is, can you give a model where no operation could recreate a theory of Z/12Z, any group, any of the popular

mathematical structures, any structures at all (i.e. one specified by finitely many things)? Is this equivalent to not having a way to make a finite set theory? maybe.

note: should replace "finite set theory" with "a theory of strings sufficient for logic"

tcridea

Try to learn some logic and see if you can describe logic and proofs as a "dynamical system" but with more than one transformation map, maybe using operads or something.

The idea is maybe we can assign a probability dist to each transformation and the consider consider a set of starting axioms A and various independent experiments $A_0^1, \ldots, A_0^n$ and for each experiment, select a random map $f_1^1, \ldots, f_1^n$ and form $A_1^1 := A_1^0 \cup f_1^1(A_0^1)$, ..., $A_1^n := A_n^0 \cup f_1^n(A_0^n)$ in parallel. Then aggregate to get $A_1 = \bigcup_j A_1^j$.

(make adjustments for correct arity of the maps).

Then repeat, i.e. set up parallel experiments, for each experiment select random function, map and, combine with previous iters, agregate, repeat.

The idea is that the aggreation will shrink the set overall list of things as much as possible, and we can match with known results to possibly find "nearby" results in the tree of computation so that we can quantify how related these randomly generated propositions are. Perhaps we can tune the prob dist so that or start the experiment in blocks so that if a given area, say geometry uses common results and definitions, we can prioritize maps and inputs that are more likely of that form of contain those as substrings, etc. Then if we can make an automated definition giver, we can collapse any long proposition by replacing substrings of known defs with those words and substrings which are not known defs into temproarily named defs and hierarchicaly structure/cluster them based on the known defs they use. The props with the most known defs and the most repeated unknown defs can be investigated first.

If we loop back to a previous result, we can remove that mapping or do something to trim and stop repeated loops.